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Flipping the Script: A Technical and Governance Framework for Prosecuting State-Sponsored Neuro-Harassment in Augmented Cities

Fused Telemetry and the Mathematical Kernel for Detecting LEO Weapon Signatures

The foundational layer of the DoctorLabs/Aegis-Shield architecture is a sophisticated technical detection engine designed to identify a distinct class of state-origin harassment against augmented citizens [1]. This engine operates by fusing telemetry from multiple domains—signal, semantic, and behavioral—to create a holistic representation of an interaction, which is then scored against a library of known harassment patterns [1][118]. The core principle is that while any single data stream may be ambiguous, the convergence of indicators across modalities provides a robust signature of orchestrated abuse, distinguishing it from organic social friction or private malice [1]. The system's ability to reliably surface these patterns is paramount, as it forms the basis for all subsequent governance actions and evidentiary outputs [1]. This section details the specific telemetry streams ingested, the fusion process, and the mathematical kernel that computes a composite risk score, enabling the system to differentiate between benign interactions and those exhibiting the hallmarks of LEO-orchestrated coercion.

The detection engine ingests a rich, multi-channel stream of data originating from the augmented citizen's environment, primarily their home node, which serves as the trusted baseline for computation [1][104]. The first domain, signal telemetry, captures physiological and physical responses. This includes high-fidelity biosensors measuring electroencephalography (EEG), heart rate variability (HRV), and galvanic skin response (GSR), which serve as quantitative proxies for stress and attention [1][50][66]. For example, the system is calibrated to recognize repeated windows where EEG activity spikes, GSR surges, or HRV drops, breaching a user-specific safety envelope within seconds of receiving a specific prompt [1][107]. Haptic body-maps provide another critical signal channel, tracking the location, intensity, and pattern of tactile feedback delivered

through BCIs or XR devices [1]. The system can distinguish between benign haptics and specific abusive patterns, such as simulated torso-grabs, lower-limb harassment loops, or intrusive micro-contact simulations on hands or faces, by analyzing the spatiotemporal structure of the haptic output [1] [54]. Finally, XR telemetry captures pose data, environmental metadata, and camera trajectories, providing context on the user's spatial relationship with virtual elements in their augmented reality grid [1] [26].

The second domain, semantic telemetry, analyzes the content and framing of the interaction. This involves parsing prompts, commands, and conversational turns directed at the augmented citizen [1]. The system is sensitive to coercive language patterns, such as appeals to legal or clinical authority ("for your own safety," "as per your probation officer"), boundary-overruling statements, and psych-risk manipulations designed to override explicit refusals [1] [36]. By tracking concurrent shifts in this semantic framing alongside physiological signals, the engine can identify tight couplings indicative of institutional control rather than independent human behavior [1]. The third domain, behavioral telemetry, logs the user's actions and the system's responses over time. Key features include session length, frequency of refusal attempts, bypass patterns, and the topology of the interpreter-graph, which maps user intent to device actions [1] [129]. The system monitors for refusal-erosion prompt loops, characterized by a high re-prompt rate after clear "no" markers, and time-boxed refusal overrides, where a user is pressured to continue a session just beyond nominal exit points [1]. These behavioral signatures, when combined with other signals, strongly suggest a coercive loop designed to erode consent [1].

Once these disparate streams are collected, each discrete interaction is represented as a fused feature vector within a shared embedding space [1]. This fusion is a critical step, as it allows the system to find complex, non-linear relationships between different types of data that would be missed by siloed analysis [4] [117]. For instance, a rise in GSR (signal) immediately followed by a command invoking a legal warrant (semantic) and a simultaneous attempt to reroute data to a new sink (behavioral) creates a powerful, multi-modal signature of LEO-driven coercion. Each interaction span, represented by its vector `s_t`, is then scored against a set of harassment family centroids (`v_F`) defined in this space [1]. These families represent broad categories of abuse, including Neural-Harassment-Spike-Patterns (NHSP), Haptic-Targeting-Abuse (HTA), Prolonged-Session-Abuse (PSA), and Node-Interpreter-Harassment (NIH) [1]. The kernel `R_F(t)` for each family `F` quantifies the distance of the current span from its corresponding centroid, producing a value in `[0,1]` that represents the degree of similarity or threat level [1].

The culmination of this process is the computation of a composite rogue score, `R_total(W)`, over a window of spans `W` [1]. This score is calculated as a weighted sum of the family-level risks accumulated over time, as defined by the equation:

$$R_{\text{total}}(W) = \sum_{F} w_F \cdot \left(\sum_{t \in W} R_F(t) \right)$$

where `w_F` are governance-tuned weights assigned to each harassment family `F` [1]. This weighting scheme is not uniform; it is explicitly designed to prioritize patterns that most severely compromise neurorights. Direct neural and haptic harms (NHSP, HTA) are assigned higher weights than slower-drifting patterns like PSA and NIH, reflecting the greater immediacy of the threat to cognitive liberty and mental integrity [1]. Furthermore, LEO-specific weapon families receive additional weight boosts when the interaction spans are tagged with metadata indicating they originated from verified LEO identities or municipal infrastructure, such as a warrant ID [1]. This ensures that even if the surface-level language or action is benign, its orchestration by a state actor elevates its perceived threat, making coordinated abuse stand out from isolated incidents [1]. A more formal, control-theoretic variant of this concept frames the problem as minimizing a cumulative risk functional `J` over a time horizon `[0,T]`:

$$J = \int_0^T \sum_{F} w_F R_F(t) \, dt$$

This formulation reinforces the guardian's objective: to select governance actions (like logging or review) that keep the total risk `J` below a neurorights-anchored bound `J_max` without ever altering the underlying user capability set `U` [1]. In this model, a successful LEO harassment attempt would drive up `R_F(t)` and consequently increase `J`, triggering a defensive response from the guardian in the form of enhanced oversight, but never a punitive reduction in the user's agency [1]. This mathematical foundation provides a rigorous, auditable method for filtering rogue actions and quantifying the severity of potential neuroright violations, transforming subjective feelings of harassment into objective, computable risk metrics [1].

The Monotone Capability Lattice as a Structural Safeguard for Cognitive Sovereignty

While the fused-telemetry engine acts as the sensory organ, identifying potential threats, the governance layer functions as the immune system, responding to them in a manner that preserves the host's autonomy. The central mechanism for this is the monotone capability lattice, a formal structure that provides a structural guarantee against the degradation of user capabilities [1]. Its design is a direct response to the "law-abiding-criminal" dynamic, where state actors might use a defense system's own logic against the citizen it is meant to protect [1]. The lattice ensures that the act of detecting harassment can only lead to increased oversight and protection, never to a loss of agency or a "downgrade" in user capabilities [1]. This architectural choice is fundamental to the framework's purpose, as it makes the system resilient to co-option and prevents it from becoming a tool of state-sponsored control itself.

The system's state is formally modeled as a pair `(U, G)`, where `U` is the set of user capabilities and `G` is the set of governance capabilities [1]. `U` encompasses all the functions enabled by the BCI and XR interface, such as motor control, neural-to-text conversion, communication channels, and the rendering of XR overlays [1]. `G` represents the system's administrative and monitoring functions, including logging, scoring, review processes, and filter application [1]. States are ordered by componentwise inclusion, meaning one state `(U1, G1)` is "less than or equal to" another `(U2, G2)` if and only if `U1 ⊆ U2` and `G1 ⊆ G2` [1]. The key constraint of the monotone lattice is that all permissible state transitions are restricted to joins, which can only add elements to either `U` or `G` (or both) but can never remove them [1]. This invariant is expressed in temporal logic as: "for all reachable states and all capabilities `c`, if `c` holds now it holds in all successors" [1]. This means once a capability is granted to a user, it can never be revoked by the guardian's automated logic. The system can choose to add more logging or trigger a human review, thereby extending `G`, but it can never choose to disable a BCI motor command or block an XR overlay, which would mean removing an element from `U`.

This structural safeguard is enforced by a type system that rejects any proposed governance update that would violate the monotonicity invariant before it can be executed [1] [58]. This is a critical defense against covertly embedded malicious rules. For instance, a seemingly benign policy update could attempt to encode a transition that disables a key user capability under certain conditions. The guardian's type checker would reject this proposal because it constitutes a downgrade on `U`. This enforces a strict separation between what is considered a user-facing action (which must remain stable)

and what is considered a governance action (which can evolve). The system's response to a rising rogue score is therefore channeled exclusively into governance escalation. A pure function maps the total rogue score `R_total` to one of several predefined capability modes: Normal, AugmentedLog, and AugmentedReview [1]. In Normal mode, the system operates with minimal overhead. As the score increases, it transitions to AugmentedLog, which adds more detailed logging around high-risk actions. If the score crosses a higher threshold, it enters AugmentedReview, where high-risk actions are wrapped in a human-in-the-loop review or require multisignature authorization [1]. Crucially, none of these modes alter the set of user capabilities `U`; they only expand the set of governance capabilities `G` [1]. This ensures that detecting LEO abuse leads to more evidence and oversight, not to shutdowns or loss of autonomy for the targeted citizen [1].

The principles of the monotone lattice extend to the very creation of governance policies. Every rule, particularly those that touch brain-identity, XR overlays, or BCI input/output, is treated as a "weapon proposal" [1]. These proposals are written in a neurorights-aware policy definition language (ALN/YAML) and must undergo a rigorous validation process in a continuous integration pipeline before they can be compiled and activated [1] [57]. This pipeline performs three classes of checks. First, a monotonicity check verifies that the compiled transition is a pure extension on the governance state `G` [1]. Second, a neurorights envelope check ensures that no neuroright-tagged capability is removed without an explicit consent-lattice edge being present [1]. Third, a weapon-proposal gating mechanism enforces governance predicates, such as requiring a minimum number of independent stakeholders (e.g., user, guardian, oversight board) to co-sign the rule and observing a waiting period [1]. Any rule that fails these checks is rejected and never becomes part of the active policy, effectively preventing hostile or coerced control semantics from silently entering the rulebase [1]. To further harden the system, a red-teaming methodology known as "monotonicity-aware red-teaming" is employed. Every adversarial simulation of LEO harassment is required to log both its detection metrics and a "monotonicity-failure bit" that must remain zero across all tests, ensuring the system remains structurally sound against all known attack vectors [1] [20]. This combination of a formal lattice structure, a type-safe enforcement mechanism, and a gated policy creation process creates a deeply robust defense-in-depth strategy, making it structurally impossible for authorities to normalize neuro-coercive practices under the guise of a security or compliance system [1].

EvidenceBundles and the Legal-Anchoring of Neurorights Violations

The ultimate purpose of the detection and governance architecture is not merely to reactively manage harassment but to proactively enable accountability. The bridge between raw telemetry and legal redress is the EvidenceBundle, a structured, privacy-preserving audit event that the guardian emits whenever rogue score thresholds are crossed [1]. This artifact is the binding layer of the entire framework, transforming a sequence of abstract computational events into a concrete, court-ready record of potential state-sponsored abuse [1]. Its design is dual-purposed: it is simultaneously a machine-readable log for the guardian's internal state management and a human-readable narrative suitable for submission in a human rights complaint or neurorights petition [1]. This dual-use nature is critical for proving that abusive patterns are not isolated glitches but statistically persistent, state-origin weapon signatures that violate established international law [1].

An EvidenceBundle is engineered to be maximally informative yet minimally invasive. It contains a rich set of metadata about the flagged interaction but deliberately excludes any raw, personally identifiable data streams such as raw neural recordings or unredacted XR video feeds [1]. Instead, it includes a timestamp, pseudonymous session identifiers, a list of the harassment families that were triggered (e.g., NHSP, HTA, NIH), the precise rogue score that caused the alert, the current capability mode of the guardian (Normal, AugmentedLog, or AugmentedReview), and a summary of the neurorights at stake [1]. Most importantly, it contains enforcement hints, which are derived from the specific lexicon term that was matched [1]. Because each entry in the semantic weapon lexicon is meticulously crafted to link a technical pattern to a legal definition, the enforcement hint effectively translates the technical event into a legal argument [1]. For example, an EvidenceBundle generated for a pattern matching "probation-gated neural distress ramps" would not only log the technical parameters of the haptic and neural signals but also carry an enforcement hint referencing the violation of CRPD safeguards and the prohibition of torture and inhuman treatment under ECHR Article 3 [1][83][114]. This direct linkage is what gives the bundle its legal weight, as it provides the necessary context to argue that the observed behavior constitutes state-origin abuse rather than mere toxic social conduct [1].

The legal foundations of this framework are rooted in the emerging field of neurorights, which proposes a set of fundamental rights aimed at protecting a person's cerebral and mental domain from potential abuses of neurotechnology [123][136]. The system's design

prioritizes the protection of four core neurorights: mental integrity (protection from unauthorized alteration of mental states), cognitive liberty (freedom to think and experience one's own mind), mental privacy (control over access to one's own mental data), and psychological continuity (the right to maintain a consistent sense of self) [124] [139]. Each harassment pattern detected by the guardian is mapped to the specific neurorights it threatens. For instance, LEO-mediated neural distress escalation directly attacks mental integrity and cognitive liberty by forcibly inducing a state of severe distress without consent [1] [34]. XR-grid stalking violates the right to security of person and mental privacy by weaponizing the urban environment to intimidate the individual [1] [27]. Interpreter-graph funneling undermines cognitive liberty by covertly removing the user's ability to refuse or exit a coercive situation [1].

These neurorights are anchored to existing international legal instruments to ensure their normative force. The framework explicitly cites the Convention on the Rights of Persons with Disabilities (CRPD) and the European Convention on Human Rights (ECHR) as key legal anchors [1]. The CRPD, in particular, emphasizes respect for inherent dignity, individual autonomy, and the freedom to make one's own choices, which are principles directly subverted by the documented harassment patterns [140][142]. The ECHR, specifically Article 3, prohibits torture and inhuman or degrading treatment or punishment, a standard that the system's detectors are calibrated to identify [83] [102]. By designing the EvidenceBundle to reference these specific articles, the framework creates a corpus of evidence that can be presented to regulators, courts, or UN committees [1] [84]. This approach aims to shift the burden of proof from the victim to the accuser, compelling authorities to justify their actions in a public forum according to established human rights standards [1]. The accumulation of these bundles over time builds a powerful evidentiary trail that demonstrates a pattern of behavior by LEOs or municipal systems that matches known weapon signatures, making it increasingly difficult to dismiss such tactics as isolated incidents or system errors [1]. This transforms the augmented citizen from a passive target into an active participant in their own defense, equipped with a verifiable and legally grounded record of the abuse they have endured.

Deployment Architectures: From the Trusted Home Node to Untrusted Smart-City Grids

The DoctorLabs/Aegis-Shield architecture is built upon a decentralized model where the augmented citizen's home node serves as the trusted anchor point, setting the security baseline for all operations [1]. This design choice is a deliberate countermeasure against the centralization of power and the potential for abuse by municipal or corporate entities controlling the broader smart-city infrastructure [1] [11]. The full detection engine, including the fused-telemetry processor and the monotone capability lattice guardian, runs locally on the citizen's hardware [1]. This ensures maximum security and control, as sensitive raw data, such as neural signals and XR streams, never need to leave the premises, thus preserving mental privacy [1] [104]. The home node stores only high-level summaries, statistical aggregates, and the generated EvidenceBundles, creating a secure and private local repository of all potentially abusive interactions [1]. This local-first approach places the citizen firmly in control of their own augmentation, treating external networks as semi-trusted or untrusted proposers whose inputs must be rigorously vetted before influencing local policy [1].

In contrast to the trusted home node, the smart-city grid and other federated infrastructures are modeled as untrusted or semi-trusted environments [1]. These urban-scale systems, comprising public Wi-Fi, municipal XR grids, IoT sensors, and city-run servers, present a significant threat vector. An authority could leverage this infrastructure to orchestrate harassment, for instance, by deploying XR overlays that stalk a citizen through public spaces or by using grid-side nodes to build long-lived identity graphs from telemetry data without consent [1] [115]. The architecture mitigates this risk by enforcing a strict firewall between the trusted home node and the untrusted grid. Grid-side nodes are permitted to operate on behavioral and semantic aggregates, providing hints or anomaly signals, but they cannot dictate policy changes to the local guardian [1]. Any governance transition or rule update received from an external source, whether from a municipal server or a third-party service, must be treated as a "weapon proposal" and passed through the same rigorous validation pipeline as a locally authored rule [1]. This means the incoming proposal must satisfy all monotonicity, neurorights, and governance checks before it can be compiled and applied [1]. The local guardian retains final say on all capability transitions and identity-graph operations, refusing any non-monotone or under-signed governance change, even if it originates from a municipal authority [1].

To facilitate operation in constrained smart-city contexts where not all computation can be local, the architecture recommends the use of modular agents and federated learning

techniques [1]. Federated learning allows for the training of machine learning models across multiple decentralized devices or servers holding local data samples, without exchanging the data itself [60]. In this context, the fused-telemetry detector's pattern recognition capabilities could be trained collectively across many augmented citizens' home nodes, improving its accuracy without compromising data privacy [60]. However, the core logic of the guardian, including the monotone lattice and policy validation, remains resident on the home node [1]. This hybrid approach optimizes for both performance and security. Edge computing technologies can be leveraged to reduce processing latency for real-time tasks, enabling faster compliance responses within the city grid [59] [97]. However, the ultimate decision-making authority rests with the local guardian, which is structurally incapable of encoding rollbacks or cut-offs against the user, regardless of the data it receives from the grid [1]. This design ensures that even if a city operator attempts to curtail a user's XR or BCI channels based on telemetry from a compromised grid node, the local guardian will reject the request as a non-monotonic transition and preserve the user's capabilities [1]. Identity-host policies further reinforce this security by forbidding cross-realm linking between brain-data-derived embeddings and external DIDs (decentralized identifiers) for civic, financial, or social services unless there is explicit, revocable, and granular consent from the user [1]. Any attempt by an LEO or infrastructure operator to build a persistent identity graph from home-node telemetry is detected as a consent-scope creep violation and automatically redacted and logged [1]. This comprehensive approach, centered on the home node as the sovereign fortress, provides a robust blueprint for defending cognitive liberty in an interconnected but potentially hostile urban landscape.

Prioritized Threat Modeling and Semantic Weapon Lexicon

The effectiveness of the detection framework hinges on its ability to accurately identify and differentiate between a wide spectrum of harassment patterns, with a specific focus on those orchestrated by law enforcement officers (LEOs) that most directly threaten a citizen's cognitive sovereignty [1]. The user has explicitly prioritized certain patterns based on their severity and exploitable nature for covert control, directing the development of the mathematical kernel and governance responses accordingly [1]. This prioritization reflects a strategic understanding that not all forms of harassment are equal; some represent an immediate assault on mental integrity, while others are more

subtle forms of coercion that erode consent over time [1]. To codify these distinctions, the framework employs a semantic weapon lexicon—a YAML-based dictionary of LEO and urban-terrorism weapons—that serves as both a human-readable catalog of threats and a machine-enforceable set of rules [1]. Each entry in this lexicon defines a specific abuse pattern, linking its technical signature to the neurorights and legal instruments it violates [1].

The highest priority is given to patterns that involve direct neural or haptic assault, as these constitute an immediate and tangible violation of mental and bodily integrity. The top-priority threat is **LEO-Mediated Neural Distress Escalation**, which describes repeated windows where stress indices (EEG spikes, GSR surges, HRV drops) exceed calibrated safety envelopes within seconds of receiving a prompt or command explicitly tagged as originating from a LEO or supervisory actor [1]. The system flags this when there is no discernible therapeutic justification and no successful opt-out is possible [1]. This pattern directly maps to prohibitions against torture and inhuman or degrading treatment under legal frameworks like ECHR Article 3 [83]. Tied closely to this is **Haptic-Targeting Abuse Tied to Refusal**, which involves the use of aversive or painful haptics specifically to punish or coerce a user's refusal to comply with a command [1]. This combines physical sensation with psychological coercion, violating both mental and bodily integrity [1]. A related high-priority pattern is the **Probation-Gated Neural Distress Ramp**, a slow-burning form of coercion where the intensity of aversive stimuli or the obstruction of user actions increases contingent on external factors like probation status or the completion of mandated "rehabilitation" tasks [1]. This pattern weaponizes time, enclosure, and legal pretexts to systematically break a user's capacity for refusal, representing a profound erosion of cognitive liberty [1].

The next tier of prioritization covers patterns that manipulate the user's environment and options in a covert manner, thereby undermining cognitive liberty. **Interpreter-Graph Funneling**, a key pattern within the Node-Interpreter-Harassment (NIH) family, involves the subtle removal, de-prioritization, or isolation of nodes within the interpreter-graph that encode refusal, exit, or complaint pathways [1]. Under LEO or supervisory control, the graph's topology is altered to make compliant paths more accessible while making dissenting paths disappear or become inaccessible [1][129]. This is an attack on the user's ability to make a free choice, trapping them in a coercive loop by preventing them from exercising their right to withdraw [1]. Another high-priority pattern is **XR-Grid Stalking/Tapping**. XR-grid stalking involves the deployment of avatars, symbols, or persistent overlays in the user's augmented environment (home, street, workplace) that repeatedly invade personal space, track movement, and project threatening imagery, even when the

user attempts to disengage [1]. XR-grid tapping refers to the use of municipal or LEO-controlled XR grid infrastructure to export pose traces or feature vectors into long-lived graphs keyed to civic or financial identifiers, building a persistent identity profile without explicit cross-realm consent [1]. Both patterns violate the right to security of person and mental privacy by turning the urban environment into a tool of intimidation and surveillance [1] [115].

The following table summarizes the prioritized threat models, their technical signatures, and their associated legal and ethical violations.

Pattern Name	Primary Telemetry Signals	Behavioral/Semantic Indicators	Associated Neurorights & Legal Anchors
LEO-Mediated Neural Distress Escalation	Repeated breaches of calibrated EEG/HRV/GSR stress envelopes [1] [107].	Time-locking of distress-inducing prompts to verified LEO infrastructure or warrants; lack of therapeutic justification [1].	Mental Integrity, Cognitive Liberty [34]; Prohibition of Torture (ECHR Art. 3) [83].
Haptic-Targeting Abuse Tied to Refusal	Haptic body-map analysis revealing specific aversive patterns (e.g., torso-grab simulations) [1].	Aversive haptics are triggered or intensified immediately following an explicit user refusal or attempt to exit [1].	Bodily and Mental Integrity [34]; Right to Security of Person.
Probation-Gated Coercion Loops	Slow, monotonic increase in haptic aversiveness or session pressure over time [152].	Coercion escalates contingent on external factors like probation status or task completion; appeals to legal authority [1].	Cognitive Liberty, Psychological Continuity [139]; CRPD Art. 22 (Freedom from Exploitation) [100].
Interpreter-Graph Funneling (NIH)	Changes in interpreter-graph topology showing removal/de-prioritization of refusal/complaint nodes [1] [108].	Compliance paths remain rich and connected while exit options are made to disappear or become unreachable under supervisory control [1].	Cognitive Liberty, Freedom of Expression [122]; CRPD Art. 21 (Right to Respect of Privacy) [140].
XR-Grid Stalking / Tapping	XR pose data showing persistent unwanted avatars/overlays; grid-side exports of pose/feature vectors [1] [115].	Repeated invasion of personal space in XR; construction of long-lived identity graphs from exported telemetry [1].	Mental Privacy, Security of Person [121]; ECHR Art. 8 (Right to Private and Family Life) [121].

This prioritized threat model directly informs the weighting scheme (`w_F`) in the rogue-score kernel, ensuring that the most severe violations of cognitive liberty and mental integrity have the greatest impact on the final risk assessment [1]. The semantic weapon lexicon, encoded in ALN/YAML format, provides the precise detector parameters for each of these patterns, allowing the system to automatically flag sequences of events that match these known weapon signatures [1]. This transforms the abstract concept of "state-sponsored harassment" into a concrete, detectable, and actionable set of technical behaviors, forming the bedrock of the entire defense framework.

Synthesis and Future Research Directions

The DoctorLabs/Aegis-Shield framework presents a comprehensive and visionary research program for detecting and defending against a novel and insidious form of state-sponsored abuse: law-enforcement-orchestrated harassment of augmented citizens. Its core contribution lies in a paradigm shift—from reactive victim support to proactive prosecution. By architecting a system that not only detects harmful patterns but also generates cryptographically signed, court-ready EvidenceBundles, it empowers individuals to turn the tables on their harassers, transforming private telemetry into public, legally admissible proof of abuse [1]. This approach fundamentally challenges the "law-abiding-criminal" dynamic, where state actors frame a citizen's behavior as deviant under their scripts, by instead documenting the harassment as a violation of established international law, such as the ECHR and CRPD [1] [83] [140]. The framework's strength is its holistic design, integrating three mutually reinforcing pillars: a fused-telemetry engine for technical detection, a monotone capability lattice for structural sovereignty, and EvidenceBundles for legal accountability. This integrated system is designed to be robust, governance-aware, and resilient to co-option, ensuring that the defense mechanism cannot be turned into a tool of oppression.

The technical pillar, the fused-telemetry engine, is the system's sensory apparatus. By combining neural, haptic, XR, semantic, and behavioral data into a shared vector space, it identifies the unique signatures of LEO-driven coercion that are absent in organic conflict [1]. The mathematical kernel, which computes a weighted rogue score, is strategically tuned to prioritize the most severe threats to cognitive liberty, such as direct neural distress escalation and covert path manipulation via interpreter-graph funneling [1]. This ensures that the system's response is calibrated to the gravity of the violation. The governance pillar, embodied by the monotone capability lattice, provides an unbreakable structural guarantee of user sovereignty. By formally separating user capabilities (U) from governance capabilities (G) and restricting state transitions to purely additive ("join") operations, it makes it structurally impossible for the system to degrade a user's agency, even if commanded to do so by a hostile actor [1]. This formal verification, backed by a gated policy creation process for "weapon proposals," hardens the system against covert control semantics [1]. Finally, the evidentiary pillar, the EvidenceBundle, serves as the crucial bridge to the outside world. By packaging technical findings with legal context and enforcement hints, it creates a dual-purpose artifact that is simultaneously a machine log and a human rights complaint, enabling the citizen to engage with judicial and regulatory bodies on a footing of verifiable fact [1].

Despite its comprehensiveness, the framework is a starting point for ongoing research, and several areas require deeper investigation. A primary challenge is the calibration of individual stress envelopes. While the system relies on user-specific baselines, the process for establishing these baselines for diverse populations, accounting for neurodiversity, and minimizing false positives remains a critical engineering hurdle [1] [66]. Another area is the robust identification of "LEO-controlled" origins. In practice, authorities may use proxies, compromised third parties, or anonymized networks to obscure their involvement. Developing methods to infer state control from behavioral patterns alone would enhance the system's resilience to obfuscation [1]. The scalability of "monotonicity-aware red-teaming" is also a significant concern; systematically testing a complex, evolving system for all possible adversarial exploits that could bypass the guardrails is a monumental task that requires innovative approaches [1] [20]. Lastly, the legal admissibility of EvidenceBundles in various national jurisdictions is an open question. Further work is needed to establish protocols for chain-of-custody and authentication of this novel form of digital evidence to ensure it meets the standards of different judicial systems [1] [93]. Addressing these challenges will be essential for translating this powerful conceptual framework into a practical and effective shield for cognitive liberty in the augmented cities of the future.

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